BARTOSZ STARZYK

JUNIOR SOFTWARE & EMBEDDED DEVELOPER

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PROFESSIONAL PROFILE

Low-level / embedded developer (C++, Rust) and 1st-year Computer Science for Embedded Systems student at AGH University of Krakow. I have commercial experience in systems optimization and QA (E2E test migration, Digital Twin work), as well as hands-on knowledge of hardware design (PCB, RP2040) and low-level software development (wgpu, OpenGL, embedded).

EXPERIENCE

Software & QA Intern *March - April 2025*

Digital Technology Poland (DTP)

- Designed and implemented a new E2E testing architecture: migrated 250 tests from Cypress to Playwright, cutting execution time from a full-day cycle (with no ability to rerun individual flaky tests) down to a few dozen minutes through multithreaded processing
- Optimized data flow in the Digital Twin module, reducing execution time by an average of 70%, and designed a regression test suite that helped catch inconsistencies before production deployment

Mobile App Developer Intern *May - June 2024*

Healthcare Facilities Team in Cieszyn

- Designed and built a Kotlin application for digital tracking of hundreds of medical equipment items, replacing outdated scanners (which didn't support certain barcode formats) and manual data entry into excel
- Gathered requirements directly from hospital staff and used them to implement barcode scanning and data editing, cutting per-item registration time from about a minute to a few seconds

IT Intern (Erasmus+, international internship) *March 2024*

VITALIS Betreuungsgesellschaft für Modellprojekte mbH, Schkeuditz, Germany

- Planned, deployed, and optimized IT services across the company's infrastructure, including installing and configuring operating systems, networks, servers, and telecommunications equipment
- Designed and deployed a SmartHome network and websites
- Worked within an international, English-speaking team on software development and testing

SKILLS

Programming languages (main): C++, Rust, Python
Programming languages (additional): Kotlin, TypeScript
Embedded & Hardware: RP2040, ESP32, ARM; I2C, SPI, UART, GPIO, DMA; prototyping, soldering, 3D modeling
Tools & Technologies: Git, CMake, Playwright (E2E), CI/CD, Blender, OpenGL, Fusion 360, Docker
Interpersonal Skills: requirements analysis & system design (UX/UI), teamwork (Agile/Scrum), technical communication, fast adaptation to new technologies
Languages: Polish - native; English – C1 (advanced)

SELECTED PROJECTS

3D Model Inspector [C++ / OpenGL]

A low-level 3D model inspector featuring a custom .obj file parser written without external libraries, optimized for fast memory allocation. Implemented a rendering pipeline with dynamic lighting (Phong/Blinn-Phong) and a GUI for real-time manipulation of transformation matrices

Raspberry Pico Console [C/C++ / Embedded / RP2040]

A handheld gaming console built on the RP2040 with an ST7735 display, built entirely from scratch: a custom PCB (2nd revision after a breadboard prototype) and a custom, modular C/C++ framework (games: Flappy Bird, a 2.5D maze) sustaining 30–120 FPS. The architecture allows new titles to be added via a single configuration entry, without touching the core engine

Lucario Engine [Rust / wgpu]

A custom, cross-platform voxel engine (with potential WASM/WebGPU compilation support). Features chunked world streaming and procedural terrain generation using multi-layered Perlin noise with domain warping. Implemented a custom renderer with a greedy meshing algorithm, cutting the vertex count per chunk (32*32*64) from ~393,000 (naive approach) down to just a few hundred to a few thousand (>99% optimization)

EDUCATION

AGH University of Krakow | od 2026

Computer Science for Embedded Systems, Bachelor's degree (in progress)

Władysław Szybiński School Complex in Cieszyn | 2021 - 2026

Programming Technician (vocational qualifications INF.03, INF.04)

- Vocational certification exam: final score 97%
- Matura (high school exam): extended-level Computer Science - 86%, bilingual English - 80%